

Building RAG Pipelines: A Three-Stage **Retrieval** Architecture with Qdrant

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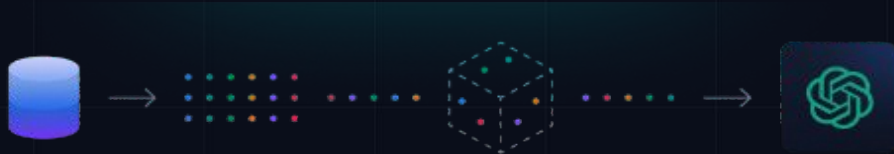
Tell me about the method used in the SpatialJB paper to jailbreak LLMs and what results did they achieve

The authors of the SpatialJB paper devise a jailbreak technique that relies on **spatially permuting the distribution of text** so that the sequence appears coherent to human readers but is interpreted as semantically incoherent by Transformer models[1]. This approach breaks the usual token-level continuity that most LLM guardrails depend on, allowing the injected content to evade detection.

In empirical evaluations, SpatialJB attains **near-perfect attack success rates (ASR)**, approaching 100% on a wide range of state-of-the-art models—including DeepSeek-V3, Llama-4, and Grok-4—even when those models incorporate advanced guardrail

What will we build ?

1. Multi-tenant Qdrant Cloud setup
2. Document ingestion
3. Three-stage retrieval architecture:
 - 3.1. Semantic search
 - 3.2. Score boosting
 - 3.3. Re-ranking with CoBERT



1. Multi-tenant Qdrant Cloud setup

Collection Creation

🔗 Multitenancy → 1 collection + Payload filtering



```
qdrant_client = AsyncQdrantClient(url=QDRANT_URL, api_key=QDRANT_API_KEY, timeout=300)
```


Collection Creation

```
await qdrant_client.create_collection(
    collection_name=COLLECTION_NAME,
    vectors_config={
        "bge_dense": models.VectorParams(
            size=768,
            distance=models.Distance.COSINE
        ),
        "colbert": models.VectorParams(
            size=128,
            distance=models.Distance.COSINE,
            multivector_config=models.MultiVectorConfig(
                comparator=models.MultiVectorComparator.MAX_SIM
            ),
            hnsw_config=models.HnswConfigDiff(m=0))},
    )
```

Named Vectors

← Semantic search

← BGE embedding dimension

← Deep re-ranking

← CoBERT token dimension

demo_collection

● GREEN

0

2

1

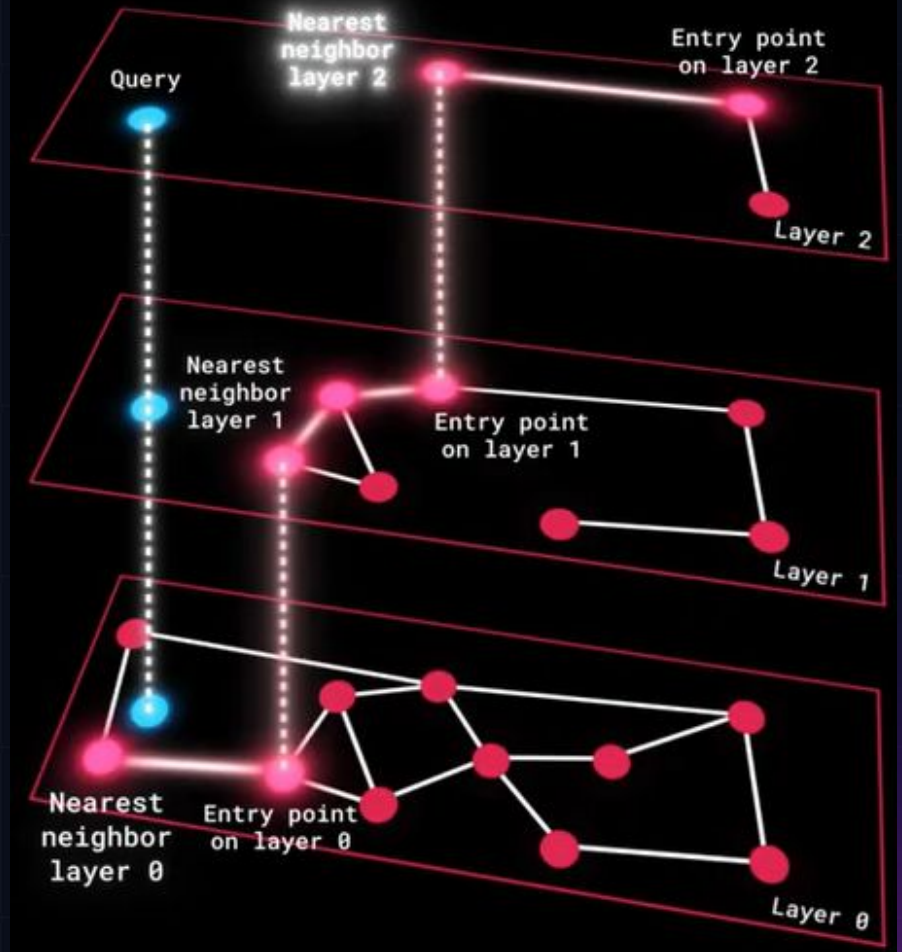
bge_dense 768 Cosine
colbert 128 Cosine

⋮

Why not index colBERT multivectors ?

- MaxSim is not symmetric $\Rightarrow$ not directly compatible with HNSW
- Used only for re-ranking at query time
- High RAM usage
- Slow insert times due to the complexity of maintaining the HNSW graph

Hierarchical Navigable Small World HNSW

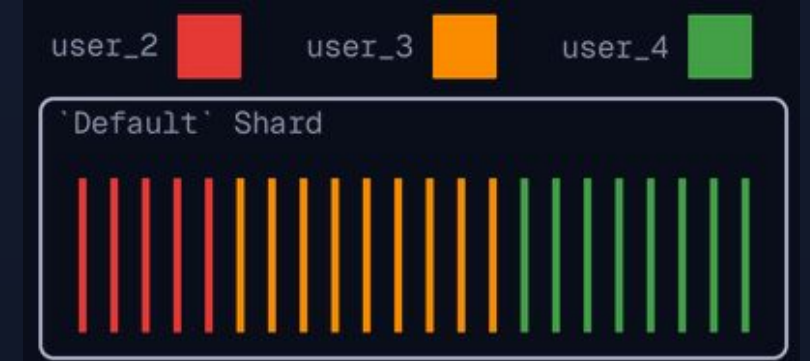


Graph-based data structure for fast approximate nearest neighbor searches

Payload Indexing

Multitenancy

```
await qdrant_client.create_payload_index(  
    collection_name=COLLECTION_NAME,  
    field_name="user_id",  
    field_schema=models.KeywordIndexParams(  
        type=models.KeywordIndexType.KEYWORD,  
        is_tenant=True,  
    ),  
)
```



Payload Indexing

Score boosting



```
other_payloads = {  
    "citation_count": models.PayloadSchemaType.INTEGER,  
    "published_year": models.PayloadSchemaType.INTEGER,  
}  
for field, schema in other_payloads.items():  
    await qdrant_client.create_payload_index(  
        collection_name=COLLECTION_NAME,  
        field_name=field,  
        field_schema=schema,  
    )
```

2. Document Ingestion

Data ingestion Steps

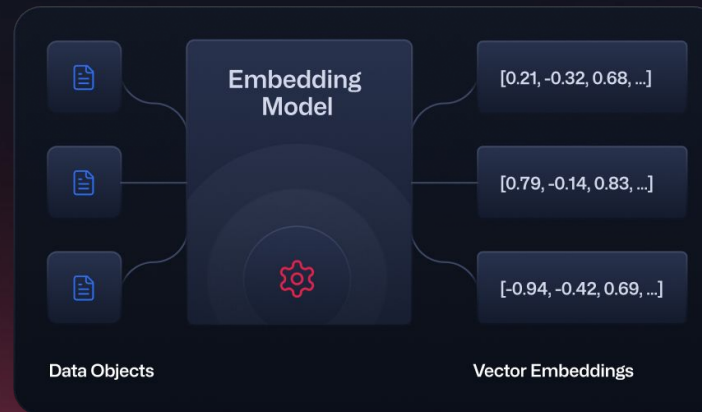
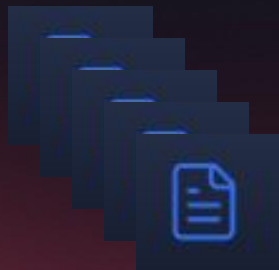
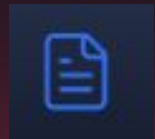
1. Chunking



2. Embedding



3. Upserting



Point 045e6c04-1fe6-413b-97e7-9664ef278918

Payload  

```
{
  "page_content": "BERT: Pre-training of Deep Bidirectional Transform...",
  "user_id": "demo_user",
  "paper_id": "1810.04805",
  "citation_count": 109088,
  "published_year": 2019
}
```

Vectors:

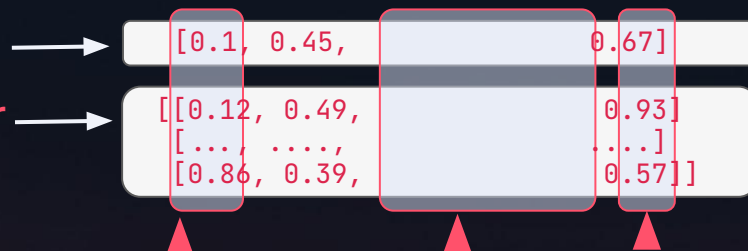
Name: colbert	Length: 282x128
Name: bge_dense	Length: 768

Embedding models

⚡ Fastembed

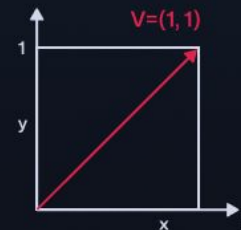
BGE vector: 768

ColBERT multivector
n_tokens x 128

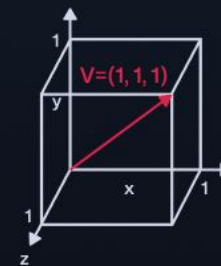


Chunk:

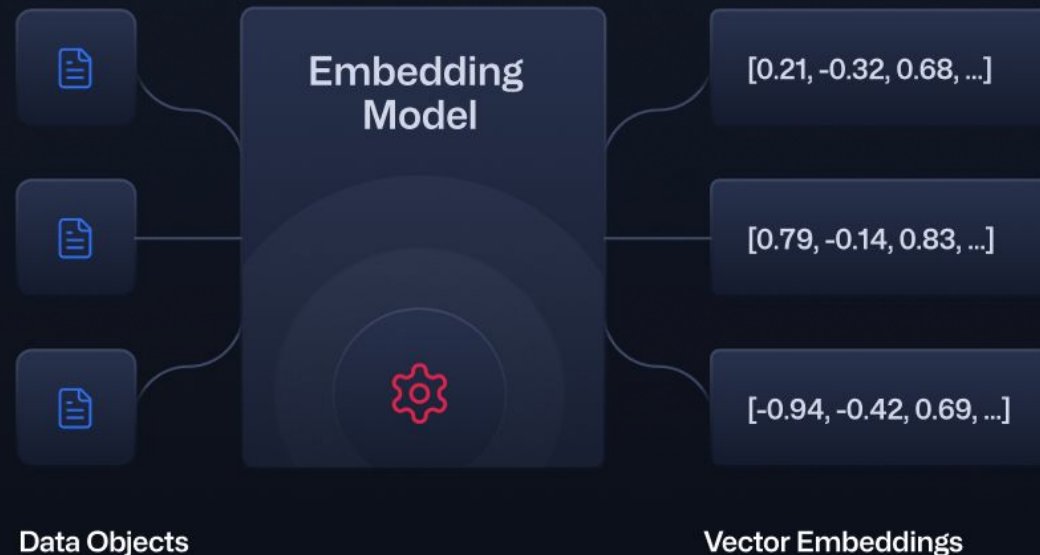
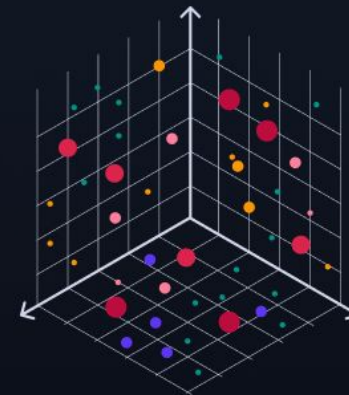
“Despite ColBERT being a powerful retriever, its speed limitation might make it less suitable for large-scale retrieval...”



2-Dimensional Vector



3-Dimensional Vector



Balancing speed and precision

BGE for scalable candidate retrieval

ColBERT for late interaction reranking.

BGE vector: 768

ColBERT multivector
n_tokens x 128



Chunk:

“Despite ColBERT being a powerful retriever, its speed limitation might make it less suitable for large-scale retrieval...”


```
dense_model = TextEmbedding("BAAI/bge-base-en-v1.5")
colbert_model = LateInteractionTextEmbedding("colbert-ir/colbertv2.0")

for i, text in enumerate(texts):
    points.append(models.PointStruct(
        id=str(uuid.uuid4()),
        vector={
            "bge_dense": dense_embeds[i].tolist(),
            "colbert": colbert_embeds[i].tolist()
        },
        payload={
            "page_content": text,
            "user_id": user_id,
            "paper_id": arxiv_id,
            "citation_count": citation_count,
            "published_year": pub_year
        }
    ))
```

```
BATCH_SIZE = 25
for i in range(0, len(points), BATCH_SIZE):
    batch = points[i : i + BATCH_SIZE]
    await qdrant_client.upsert(
        collection_name=COLLECTION_NAME,
        points=batch,
        wait=True
    )
```

Point 045e6c04-1fe6-413b-97e7-9664ef278918

Payload  

```
{
  "page_content": "BERT: Pre-training of Deep Bidirectional Transform ...",
  "user_id": "demo_user",
  "paper_id": "1810.04805",
  "citation_count": 109088,
  "published_year": 2019
}
```

Vectors:

Name:	colbert	Length:	282x128
Name:	bge_dense	Length:	768

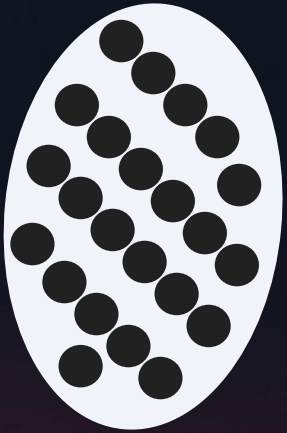
3. Multi-stage retrieval

Multi-stage Query





Filtering
by user_id



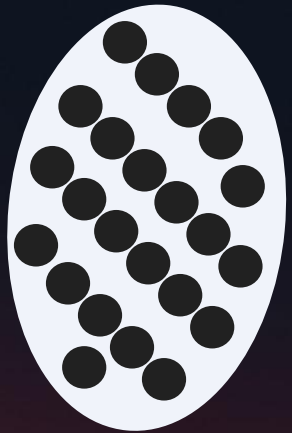
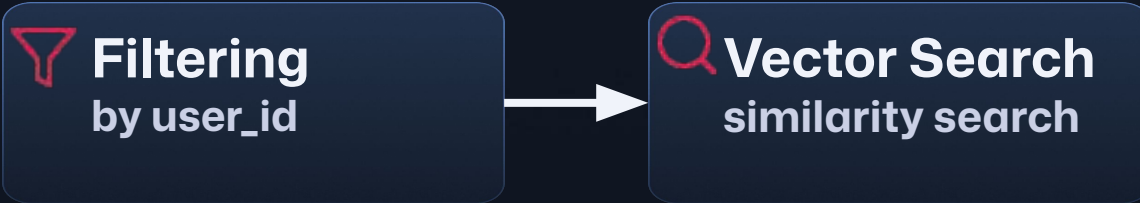
User data

1. Embedding the user query

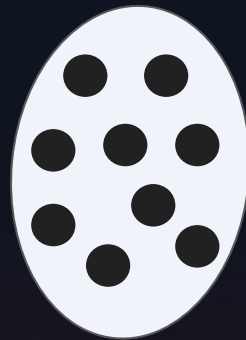
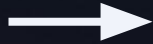
```
query_dense = list(dense_model.embed([query_text]))[0].tolist()
query_multivector =
list(colbert_model.embed([query_text]))[0].tolist()
```

2. Filtering by user_id

```
response = await qdrant_client.query_points(
    collection_name=COLLECTION_NAME,
    query_filter=models.Filter(
        must=[models.FieldCondition(
            key="user_id",
            match=models.MatchValue(value=user_id)
        )]
    ),
```



User data



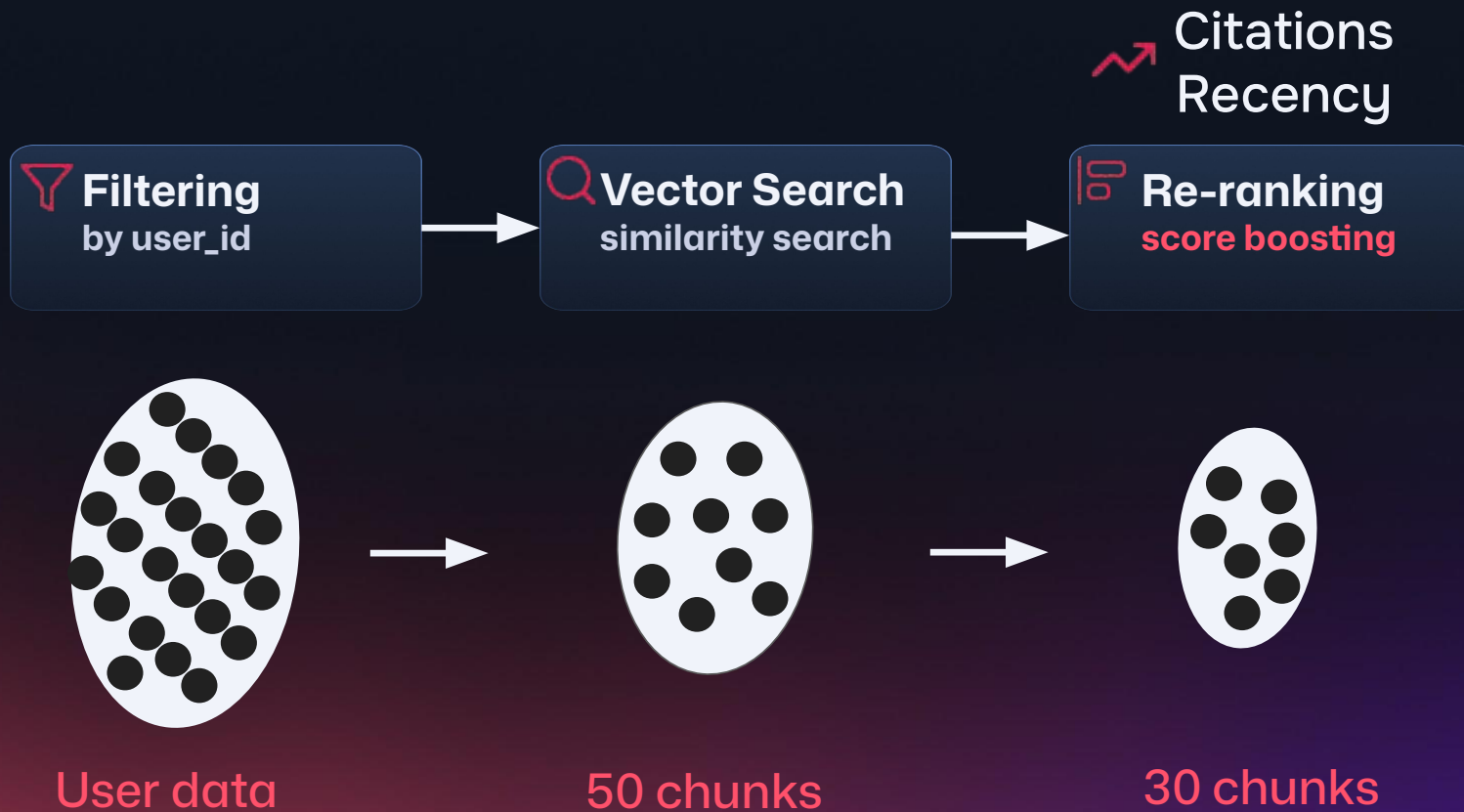
50 chunks

3. Semantic Search : Prefetch

```
prefetch=[
models.Prefetch(
    query=query_dense,
    using="bge_dense",
    limit=50
),
```


4. Score boosting with FormulaQuery

$$\text{Boosted Score} = \text{Similarity Score} + 0.1 \ln(\text{citations} + 1) + 0.05(\text{published year} - 2000)$$



```
prefetch=[
```

```
models.Prefetch(
```

```
    prefetch=[
```

```
        models.Prefetch(
```

```
            query=query_dense,
```

```
            using="bge_dense",
```

```
            limit=50,)],
```

```
    query=models.FormulaQuery(
```

```
        formula=models.SumExpression(
```

```
            sum=[
```

```
                "$score", ← Similarity score from the semantic search step
```

```
                models.MultExpression(mult=[
```

```
                    0.1,
```

```
                    models.LnExpression(ln=models.SumExpression(sum=[1.0, "citation_count"])),
```

```
                ]),
```

```
                models.MultExpression(mult=[
```

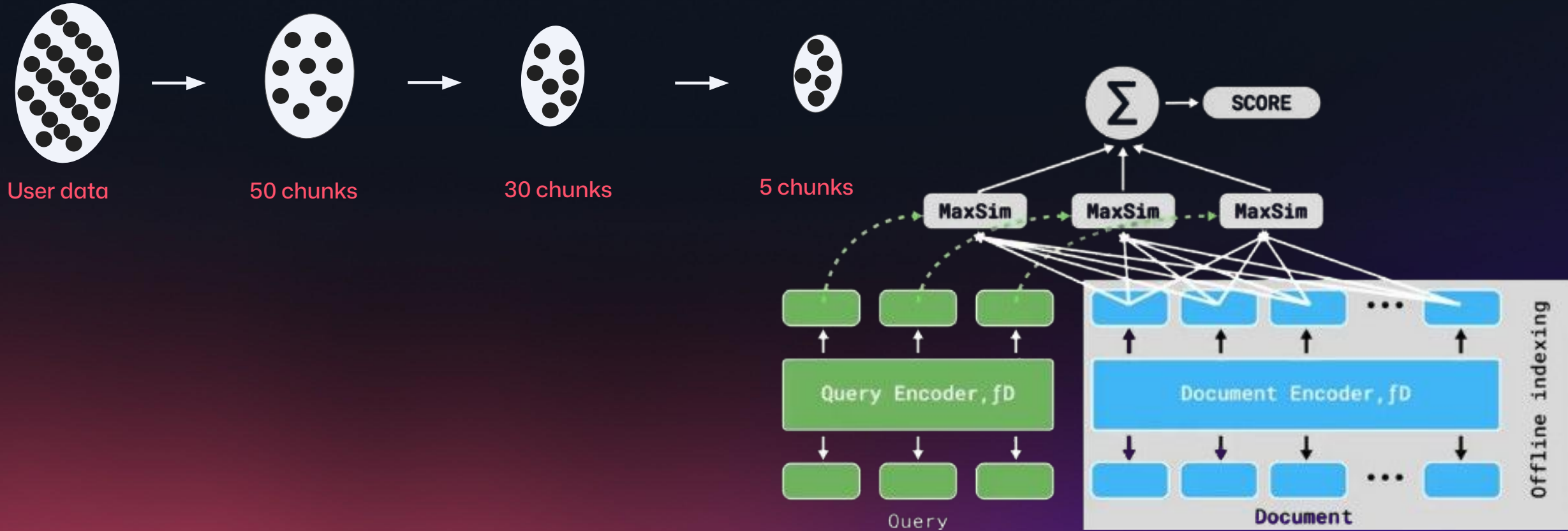
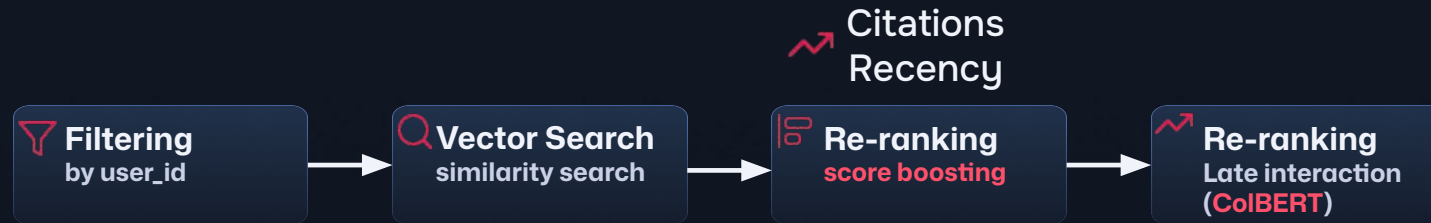
```
                    0.05,
```

```
                    models.SumExpression(sum=["published_year", -2000.0])))])),
```

```
    limit=30,)],
```

Nested prefetches

5. Late interaction re-ranking with colBERT



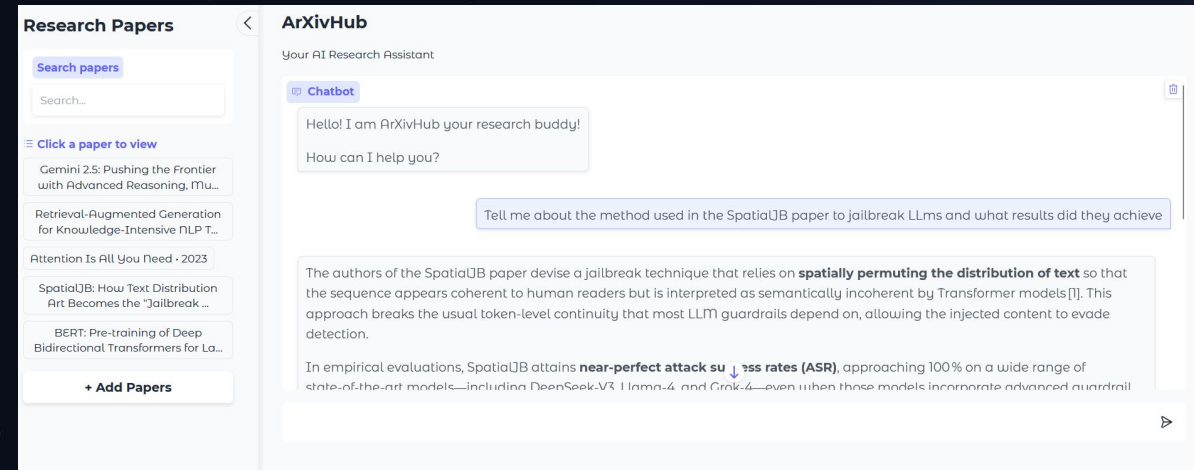
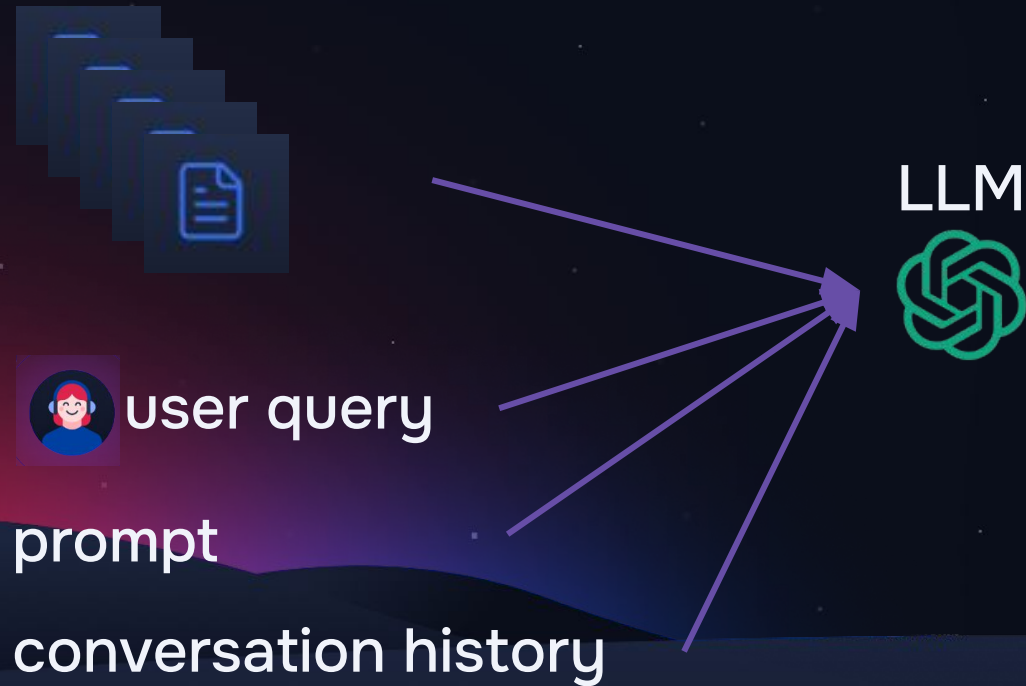
```

prefetch=[models.Prefetch(
    prefetch=[
        models.Prefetch(
            query=query_dense,
            using="bge_dense",
            limit=50)],
    query=models.FormulaQuery(
        formula=models.SumExpression(
            sum=["$score",
                models.MultExpression(mult=[0.1,
                    models.LnExpression(ln=models.SumExpression(sum=[1.0, "citation_count"]))
                ]),
                models.MultExpression(mult=[0.005,
                    models.SumExpression(sum=["published_year", -2000.0])
                ])
            ]))],
    limit=30,)],
    query=query_multivector,
    using="colbert",
    limit=5,
    with_payload=True
)
return response.points

```

Putting it all together

- Balance between scalability, latency and quality
- Tune parameter choices to your specific use case



Thank you Questions ?

